

Year 7 · Chapter 12 · Ratios, rates and time: answers (teacher copy)

12.01 Ratios (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: 3 : 5
- Ex 1b: 4 : 6
- Ex 2a: 38 : 17
- Ex 2b: 17 : 55
- Ex 3a: 2 : 1
- Ex 3b: No: the order matters

Easy (drill)

- | | | |
|-----------|------------|-----------|
| 1. 2 : 2 | 11. 5 : 18 | 21. 8 : 1 |
| 2. 1 : 7 | 12. 5 : 5 | 22. 2 : 5 |
| 3. 4 : 15 | 13. 7 : 6 | 23. 5 : 4 |
| 4. 5 : 10 | 14. 4 : 1 | 24. 1 : 9 |
| 5. 5 : 5 | 15. 2 : 3 | 25. 1 : 3 |
| 6. 8 : 10 | 16. 4 : 9 | 26. 6 : 7 |
| 7. 3 : 15 | 17. 2 : 7 | 27. 1 : 1 |
| 8. 15 : 1 | 18. 5 : 1 | 28. 6 : 5 |
| 9. 7 : 15 | 19. 1 : 1 | |
| 10. 3 : 8 | 20. 8 : 9 | |

Medium (drill)

- | | | |
|------------|------------|---------------|
| 1. 2 : 5 | 7. 2 : 15 | 13. 5 : 1 : 6 |
| 2. 2 : 3 | 8. 1 : 3 | 14. 1 : 1 : 3 |
| 3. 1 : 3 | 9. 9 : 4 | 15. 4 : 2 : 5 |
| 4. 1 : 6 | 10. 1 : 30 | 16. 2 : 2 : 3 |
| 5. 7 : 6 | 11. 1 : 3 | 17. 5 : 5 : 3 |
| 6. 1 : 200 | 12. 1 : 4 | 18. 1 : 1 : 2 |

Challenging

- 1/5
- 8
- 3 shaded, 2 unshaded in each 5 (6 : 4)

Extension

- 3/10
- 30%

Exit ticket

- Level 1: 6 : 5
- Level 2: 5 : 11
- Level 3: 6 : 7

12.02 Equivalent and simplified ratios

(booklet p. 7)

WE DO examples (teacher)

- Ex 1a: 8 : 12
- Ex 1b: 6 : 9 : 12
- Ex 2a: 4 : 7
- Ex 2b: 3 : 1 : 4
- Ex 3a: 2 : 1
- Ex 3b: 1 : 3

Easy (drill)

- | | | |
|------------|------------|--------|
| 1. 7 : 8 | 10. 1 : 11 | 19. 64 |
| 2. 2 : 9 | 11. 5 : 9 | 20. 48 |
| 3. 11 : 3 | 12. 11 : 1 | 21. 14 |
| 4. 3 : 7 | 13. 6 : 7 | 22. 63 |
| 5. 5 : 4 | 14. 5 : 12 | 23. 3 |
| 6. 11 : 7 | 15. 2 : 1 | 24. 32 |
| 7. 7 : 5 | 16. 2 : 9 | 25. 28 |
| 8. 12 : 11 | 17. 8 | 26. 16 |
| 9. 1 : 5 | 18. 18 | 27. 2 |

28. 24

29. 24

30. 7

31. 8

32. 18

Medium (drill)

- | | | |
|----------|-----------|---------------|
| 1. 3 : 7 | 9. 9 : 8 | 17. 1 : 1 : 4 |
| 2. 9 : 4 | 10. 5 : 8 | 18. 1 : 2 : 1 |
| 3. 5 : 1 | 11. 9 : 2 | 19. 3 : 2 : 4 |
| 4. 5 : 6 | 12. 4 : 1 | 20. 1 : 7 : 1 |
| 5. 2 : 5 | 13. 2 : 1 | 21. 6 : 1 : 1 |
| 6. 2 : 1 | 14. 2 : 1 | 22. 1 : 5 : 6 |
| 7. 3 : 2 | 15. 7 : 5 | |
| 8. 8 : 3 | 16. 6 : 1 | |

Challenging

- 1 : 3
- 4 : 7
- 48 : 5

Extension

- 6 : 5
- 1 : 3 : 6

Exit ticket

- Level 1: 21
- Level 2: 2 : 5
- Level 3: 3 : 20

12.03 Ratio problems (booklet p. 11)

WE DO examples (teacher)

- Ex 1a: 100
- Ex 1b: \$45 000
- Ex 2a: \$20 and \$40
- Ex 2b: \$200 and \$250
- Ex 3a: 1.5 km
- Ex 3b: 12 kg

Easy (drill)

- | | | |
|------------------|-------------------|--------|
| 1. \$20 and \$15 | 8. \$6 and \$24 | 15. 27 |
| 2. \$44 and \$11 | 9. \$24 and \$32 | 16. 16 |
| 3. \$54 and \$36 | 10. \$16 and \$8 | 17. 32 |
| 4. \$3 and \$9 | 11. \$80 and \$40 | 18. 18 |
| 5. \$15 and \$75 | 12. \$15 and \$45 | 19. 32 |
| 6. \$16 and \$4 | 13. 28 | 20. 6 |
| 7. \$33 and \$11 | 14. 8 | |

Medium (drill)

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|------------------------|-------------------------|----------------------|
| 1. 24 kg, 18 kg, 18 kg | 7. 7 m, 28 m, 14 m | 12. 20 m, 20 m, 10 m |
| 2. 6 L, 4 L, 2 L | 8. 32 L, 24 L, 8 L | 13. \$44 and \$77 |
| 3. 14 L, 28 L, 21 L | 9. 11 m, 44 m, 22 m | 14. \$6 and \$12 |
| 4. 6 kg, 6 kg, 3 kg | 10. 36 kg, 18 kg, 9 kg | 15. \$48 and \$60 |
| 5. 32 L, 40 L, 8 L | 11. 28 kg, 14 kg, 21 kg | 16. \$20 and \$25 |
| 6. 36 kg, 24 kg, 12 kg | | |

Challenging

- 320
- Sam \$36, Jo \$60
- 750 g

Extension

- 12
- A \$28, B \$14, C \$42

Exit ticket

- Level 1: 35

2. Level 2: \$30 and \$50
3. Level 3: 1 L

12.04 Rates (booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: 3000 voters/politician
2. Ex 1b: 200 sheep/ha
3. Ex 2a: \$0.80/ticket (or 1.25 tickets/\$)
4. Ex 2b: 270 stitches/ball
5. Ex 3a: 80 km/h
6. Ex 3b: 1500 m/min

Easy (drill)

- | | | |
|-----------------|------------------|---------------|
| 1. \$55/kg | 9. 19 m/s | 17. c/L |
| 2. 18 words/min | 10. 54 km/h | 18. GB/month |
| 3. 3 words/min | 11. 38 L/min | 19. L/min |
| 4. 26 beats/min | 12. 60 beats/min | 20. \$/kg |
| 5. 7 L/min | 13. words/min | 21. mm/day |
| 6. 51 words/min | 14. m/s | 22. L/100 km |
| 7. \$12/kg | 15. \$/h | 23. beats/min |
| 8. 6 km/h | 16. cm/week | 24. km/h |

Medium (drill)

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|---------------|--------------------------|--------------------------|
| 1. 0.1 km/min | 8. 5400 km/h | 14. A (44 and 32.6 km/h) |
| 2. 2 km/min | 9. 0.6 km/min | 15. B (36 and 63.3 km/h) |
| 3. 7200 km/h | 10. 3600 km/h | 16. A (53 and 36.3 km/h) |
| 4. 1800 km/h | 11. 0.6 c/g | |
| 5. 3.6 c/g | 12. 1.5 km/min | |
| 6. 0.3 km/min | 13. B (47 and 98.5 km/h) | |
| 7. 6 c/g | | |

Challenging

1. 1500 m/min
2. 18 km/h
3. 30 c/min

Extension

1. 54 km/h
2. 10.8 m/h

Exit ticket

1. Level 1: \$9/kg
2. Level 2: 75 km/h
3. Level 3: 1000 m/min

12.05 Best buys (booklet p. 19)

WE DO examples (teacher)

1. Ex 1a: \$1.30
2. Ex 1b: \$1.11; Chow
3. Ex 2a: Dairy Farm (\$2.85/L vs \$3.12/L)
4. Ex 2b: 3 L (\$2.20/L vs \$2.30/L)
5. Ex 3a: e.g. it may go off or they cannot store it
6. Ex 3b: 5 for \$6.50 (\$1.30 vs \$1.40)

Easy (drill)

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|-----------|-------------------|-------------------|
| 1. \$3.74 | 8. \$3.81 | 15. 2 for \$4.68 |
| 2. \$8.75 | 9. \$8.63 | 16. 2 for \$5.70 |
| 3. \$2.23 | 10. \$6.21 | 17. 5 for \$12.75 |
| 4. \$7.06 | 11. \$8.56 | 18. 6 for \$6.84 |
| 5. \$7.04 | 12. \$3.78 | 19. 3 for \$9.06 |
| 6. \$1.11 | 13. 4 for \$15.56 | 20. 4 for \$4.08 |
| 7. \$6.34 | 14. 5 for \$5.10 | |

Medium (drill)

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|--|--|--|
| 1. 750 g (\$1.45 and \$0.76 per 100 g) | 2. 600 g (\$3.97 and \$1.55 per 100 g) | 3. 375 g (\$0.77 and \$0.81 per 100 g) |
|--|--|--|

- | | | |
|--|---|-------------|
| 4. 250 g (\$0.67 and \$1.88 per 100 g) | 7. 600 g (\$3.78 and \$1.90 per 100 g) | 11. \$15.36 |
| 5. 600 g (\$1.27 and \$1.19 per 100 g) | 8. 1000 g (\$1.22 and \$0.75 per 100 g) | 12. \$6.80 |
| 6. 250 g (\$1.20 and \$1.51 per 100 g) | 9. \$2.68 | 13. \$46.38 |
| | 10. \$6.63 | 14. \$44.85 |

Challenging

1. 5 kg (\$2.10/kg)
2. 5 c per 100 g
3. no: waste, storage, use-by dates

Extension

1. 1 L: \$1.35 vs \$1.50 per 100 mL, 15 c cheaper
2. \$3.00

Exit ticket

1. Level 1: \$4.20/kg
2. Level 2: 2 L (\$2.70/L)
3. Level 3: 1 kg (\$0.90/100 g)

12.06 Rate problems (booklet p. 23)

WE DO examples (teacher)

1. Ex 1a: \$853.10
2. Ex 1b: 36.3 L
3. Ex 2a: 7.5 ha
4. Ex 2b: 200 km
5. Ex 3a: 3 h
6. Ex 3b: 17.28 L

Easy (drill)

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|-----------|-----------|---------|
| 1. \$150 | 11. \$12 | 21. 1 h |
| 2. \$36 | 12. \$144 | 22. 3 h |
| 3. \$209 | 13. 5 h | 23. 6 h |
| 4. 280 km | 14. 4 h | 24. 4 h |
| 5. \$75 | 15. 2 h | 25. 2 h |
| 6. \$112 | 16. 5 h | 26. 3 h |
| 7. \$138 | 17. 2 h | 27. 5 h |
| 8. 945 km | 18. 3 h | 28. 6 h |
| 9. \$80 | 19. 3 h | |
| 10. \$22 | 20. 1 h | |

Medium (drill)

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|------------|------------|------------|
| 1. 30 km | 6. 40 km | 11. 60 km |
| 2. 22.5 km | 7. 45 km | 12. 225 km |
| 3. 90 km | 8. 300 km | 13. \$870 |
| 4. 50 km | 9. 135 km | 14. \$600 |
| 5. 67.5 km | 10. 120 km | 15. \$390 |

Challenging

1. 30 min
2. \$66.60
3. 10 min

Extension

1. 360 km
2. after 2 hours

Exit ticket

1. Level 1: \$147
2. Level 2: 3 h
3. Level 3: 31.25 L

12.07 Travel graphs (booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: 22 km
2. Ex 1b: 2
3. Ex 2a: 3 p.m.
4. Ex 2b: 1 p.m. to 2 p.m.
5. Ex 3a: 4 km/h
6. Ex 3b: faster

Easy (drill)

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|-------------|------------|----------------------|
| 1. 15 km/h | 5. 80 km/h | 9. 5 h |
| 2. 100 km/h | 6. 40 km/h | 10. from 2 h to 3 h |
| 3. 10 km/h | 7. 40 km | 11. hour 1 (40 km/h) |
| 4. 45 km/h | 8. 120 km | 12. 24 km/h |

Medium (drill)

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|--------------------|---------------------|----------------------|
| 1. 20 km | 5. hour 3 (40 km/h) | 9. 6 h |
| 2. 110 km | 6. 18.3 km/h | 10. from 1 h to 2 h |
| 3. 6 h | 7. 20 km | 11. hour 3 (40 km/h) |
| 4. from 1 h to 2 h | 8. 110 km | 12. 18.3 km/h |

Challenging

1. stopped (not moving)
2. returning to the start
3. e.g. drove 160 km, stopped for lunch for 1 h, then drove on

Extension

1. graph
2. 10 a.m.

Exit ticket

1. Level 1: stopped
2. Level 2: 15 km/h
3. Level 3: 3 km

12.08 Time (booklet p. 33)

WE DO examples (teacher)

1. Ex 1a: 10 h
2. Ex 1b: 3 h; 12 min
3. Ex 2a: 5 h 16 min
4. Ex 2b: 3 min 28 s
5. Ex 3a: 10 h 5 min
6. Ex 3b: 29 min

Easy (drill)

- | | | |
|---------------|----------------|----------------|
| 1. 120 min | 9. 4 h 10 min | 17. 4 h 9 min |
| 2. 360 s | 10. 180 s | 18. 2 h 15 min |
| 3. 6 h 20 min | 11. 7 h 30 min | 19. 2 h 25 min |
| 4. 8 h | 12. 300 min | 20. 5 h 35 min |
| 5. 5 h | 13. 3 h 2 min | 21. 4 h 1 min |
| 6. 240 s | 14. 1 h 36 min | 22. 2 h 53 min |
| 7. 540 s | 15. 1 h 56 min | 23. 2 h 32 min |
| 8. 480 min | 16. 2 h 45 min | 24. 4 h 55 min |

Medium (drill)

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|---------------|----------------|----------------|
| 1. 31 min | 8. 42 min | 15. 6 h 45 min |
| 2. 2 h 58 min | 9. 18 min | 16. 5 h 30 min |
| 3. 51 min | 10. 1 h 10 min | 17. 6 h 24 min |
| 4. 1 h 47 min | 11. 3 h 2 min | 18. 3 h 12 min |
| 5. 1 h 36 min | 12. 1 h 54 min | 19. 5.2 h |
| 6. 1 h 43 min | 13. 6.4 h | 20. 6 h 30 min |
| 7. 46 min | 14. 4 h 12 min | |

Challenging

1. 9:53 p.m.
2. 10 min 55 s
3. 15 s

Extension

1. age $\times$ 8760 (approx.)
2. 11 days 13 h 46 min 40 s

Exit ticket

1. Level 1: 2 h 30 min
2. Level 2: 4 h 15 min
3. Level 3: 2 h 25 min

12.09 12-hour and 24-hour time (booklet p. 37)

WE DO examples (teacher)

1. Ex 1a: 3:50 p.m.
2. Ex 1b: 1:48 a.m.
3. Ex 2a: 12:30 p.m.
4. Ex 2b: 14:10
5. Ex 3a: 00:19
6. Ex 3b: no confusion between a.m. and p.m.

Easy (drill)

- | | | |
|-----------|----------------|----------------|
| 1. 23:05 | 12. 02:25 | 23. 12:10 p.m. |
| 2. 19:45 | 13. 14:40 | 24. 12:35 a.m. |
| 3. 19:15 | 14. 00:50 | 25. 6:50 p.m. |
| 4. 08:05 | 15. 04:50 | 26. 6:30 a.m. |
| 5. 12:50 | 16. 00:20 | 27. 7:20 p.m. |
| 6. 14:10 | 17. 7:15 p.m. | 28. 10:50 p.m. |
| 7. 03:50 | 18. 2:20 p.m. | 29. 3:45 a.m. |
| 8. 07:25 | 19. 5:55 a.m. | 30. 4:15 a.m. |
| 9. 09:00 | 20. 8:50 a.m. | 31. 3:20 a.m. |
| 10. 04:35 | 21. 8:45 a.m. | 32. 8:10 a.m. |
| 11. 03:00 | 22. 10:30 p.m. | |

Medium (drill)

- | | | |
|---------------|----------------|-----------|
| 1. 12:20 a.m. | 9. 12:30 a.m. | 17. 08:10 |
| 2. 11:10 | 10. 00:30 | 18. 14:30 |
| 3. 11:15 a.m. | 11. 12:10 p.m. | 19. 02:20 |
| 4. 00:25 | 12. 11:10 p.m. | 20. 08:15 |
| 5. 23:55 | 13. 00:05 | 21. 07:40 |
| 6. 00:45 | 14. 11:25 | 22. 10:55 |
| 7. 11:40 a.m. | 15. 00:15 | |
| 8. 11:55 | 16. 23:00 | |

Challenging

1. 02:15, 2:15 a.m.
2. 7 h 45 min
3. 23:40

Extension

1. 05:35
2. 515 min

Exit ticket

1. Level 1: 5:25 p.m.
2. Level 2: 20:05
3. Level 3: 00:45

12.10 Time differences (booklet p. 41)

WE DO examples (teacher)

1. Ex 1a: 4 h 20 min
2. Ex 1b: 7 h 35 min
3. Ex 2a: 6 h 30 min
4. Ex 2b: 30 min
5. Ex 3a: 19 years 2 months 13 days
6. Ex 3b: 18 days

Easy (drill)

- | | | |
|---------------|---------------|----------------|
| 1. 8 h | 5. 3 h 50 min | 9. 2 h |
| 2. 4 h 45 min | 6. 7 h 45 min | 10. 5 h 25 min |
| 3. 1 h 25 min | 7. 4 h 25 min | 11. 3 h 20 min |
| 4. 6 h 5 min | 8. 4 h 35 min | 12. 5 h 20 min |

- | | | |
|----------------|----------------|----------------|
| 13. 1 h 50 min | 17. 30 min | 21. 25 min |
| 14. 5 h 10 min | 18. 1 h 5 min | 22. 2 h 20 min |
| 15. 5 h 10 min | 19. 4 h 5 min | 23. 4 h 5 min |
| 16. 2 h | 20. 1 h 15 min | 24. 35 min |

Medium (drill)

- | | | |
|---------------|----------------|----------------|
| 1. 2 h 50 min | 7. 4 h 10 min | 13. 2:00 p.m. |
| 2. 1 h 15 min | 8. 5 h 5 min | 14. 10:45 a.m. |
| 3. 4 h 55 min | 9. 6 h 25 min | 15. 1:45 p.m. |
| 4. 2 h 35 min | 10. 2 h 15 min | 16. 8:40 p.m. |
| 5. 5 h 50 min | 11. 4 h 5 min | 17. 9:55 a.m. |
| 6. 3 h 30 min | 12. 4 h 25 min | 18. 6:10 a.m. |

Challenging

- 4 h 55 min
- 33 days
- 8 years 10 months

Extension

- 10:00 a.m.
- 07:00 the next day (Sydney)

Exit ticket

- Level 1: 2 h 45 min
- Level 2: 2 h 40 min
- Level 3: 15 days

12.11 Timetables (booklet p. 45)

WE DO examples (teacher)

- Ex 1a: 8:33 a.m.
- Ex 1b: 2 h 16 min
- Ex 2a: 7:40 a.m.

- Ex 2b: 1 h 13 min
- Ex 3a: the 7:40 a.m. train (arrives 9:56 a.m.)
- Ex 3b: no confusion between a.m. and p.m.

Easy (drill)

- | | | |
|---------------|----------------|------------|
| 1. 2 h 5 min | 8. 50 min | 15. 09:42 |
| 2. 3 h 20 min | 9. 2 h 30 min | 16. 09:57 |
| 3. 1 h 10 min | 10. 2 h 40 min | 17. 42 min |
| 4. 15 min | 11. 1 h | 18. 21 min |
| 5. 1 h | 12. 1 h 15 min | 19. 23 min |
| 6. 1 h 45 min | 13. 08:19 | 20. 19 min |
| 7. 35 min | 14. 09:03 | |

Medium (drill)

- | | | |
|---------------|----------------|----------------|
| 1. 2 h 15 min | 6. 55 min | 11. 2 h 25 min |
| 2. 2 h 25 min | 7. 1 h 35 min | 12. 1 h |
| 3. 1 h 55 min | 8. 1 h | 13. 50 min |
| 4. 2 h 30 min | 9. 2 h 5 min | 14. 30 min |
| 5. 20 min | 10. 1 h 30 min | 15. 2 h 5 min |

Challenging

- 06:50 → 07:15, 07:30 → 07:55, 08:10 → 08:35, 08:50 → 09:15, 09:30 → 09:55
- own timetable

Extension

- 8:33 a.m.
- 10:08 a.m.

Exit ticket

- Level 1: 10:30 a.m.
- Level 2: 21 min
- Level 3: the 6:05 a.m. train